



Micronutrient Status of Soil: A Case Study in District Pulwama, Jammu & Kashmir

This study focuses on assessing the availability and distribution of essential soil micronutrients in Pulwama District of Jammu & Kashmir, a region where agriculture forms the backbone of the local economy.

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Introduction & Background

Soil is one of the most vital natural resources supporting agricultural productivity and environmental sustainability. While macronutrients such as nitrogen, phosphorus, and potassium are required in large quantities, micronutrients are equally important despite being required in smaller amounts.

Micronutrients such as zinc, iron, manganese, copper, boron, and molybdenum play crucial roles in plant metabolic activities, enzyme activation, chlorophyll synthesis, and overall plant development. Deficiencies in these nutrients can lead to reduced crop yield, poor crop quality, and long-term soil degradation.

In Pulwama district, increasing agricultural intensification and excessive use of chemical fertilizers have disturbed the natural nutrient balance of soils, making the assessment of micronutrient status essential for sustainable agriculture.

Importance of Micronutrients in Soil Health

Micronutrients are indispensable for maintaining soil fertility and crop productivity. Each micronutrient performs specific physiological functions in plants:

- **Zinc (Zn):** Required for enzyme activity, protein synthesis, and growth regulation
- **Iron (Fe):** Essential for chlorophyll formation and photosynthesis
- **Manganese (Mn):** Involved in metabolic reactions and nitrogen assimilation
- **Copper (Cu):** Important for reproductive growth and disease resistance
- **Boron (B):** Plays a key role in cell wall formation, flowering, and fruit development
- **Molybdenum (Mo):** Necessary for nitrogen fixation and nitrate reduction

Even slight deficiencies can severely affect plant growth, yield, and nutritional quality, ultimately impacting food security and farmer livelihoods.



Study Area: Pulwama District

Pulwama district is located in the southern part of the Kashmir Valley and is widely recognized for its agricultural importance. The region experiences a temperate climate with cold winters and moderate summers, which strongly influence soil characteristics and nutrient dynamics.

The district lies between **33°34'N–33°98'N latitude** and **74°58'E–75°28'E longitude**. Major crops grown include rice, saffron, vegetables, and fruit orchards. The soils range from alluvial soils in low-lying areas to loamy and clay soils in upland and terraced regions.

Due to variations in soil type, cropping pattern, and management practices, Pulwama provides an ideal setting to study spatial variability in soil micronutrient availability.

Objectives of the Study

The study was conducted with the following specific objectives:

1 Assess Micronutrient Status

To assess the present status of essential micronutrients in the soils of Pulwama district

2 Identify Deficiencies

To identify areas affected by micronutrient deficiencies or imbalances

3 Examine Relationships

To examine the relationship between soil properties (pH, organic carbon) and micronutrient availability

4 Evaluate Awareness

To evaluate farmer awareness and nutrient management practices

5 Suggest Strategies

To suggest suitable soil management strategies for improving soil fertility and crop productivity

Research Methodology

A systematic and scientific approach was adopted for the study. A descriptive and exploratory research design was used to analyze soil micronutrient status and farmer practices.

Composite soil samples were collected using stratified random sampling from different land-use systems. Laboratory analysis was carried out to determine micronutrient concentrations. Primary data were collected through questionnaires, interviews, and field observations, while secondary data were obtained from published research and government reports.

Statistical analysis and GIS-based mapping techniques were used to interpret spatial patterns and relationships between soil parameters and nutrient availability.



Soil Physicochemical Characteristics

Soil pH

Ranged from slightly acidic to moderately alkaline

Electrical Conductivity

Indicated non-saline soil conditions

Organic Carbon

Low to medium levels, higher where organic manures were used

Available NPK

Medium to adequate fertility status

These parameters strongly influence micronutrient availability. Higher soil pH was associated with reduced availability of zinc and boron, while organic matter improved overall nutrient retention.

Results and Major Findings

The analysis revealed significant variability in micronutrient status across the district:



Zinc (Zn)

46% of soil samples were deficient, especially in intensively cultivated paddy fields



Boron (B)

38% deficiency, mainly in orchard soils with coarse texture



Iron (Fe)

Mostly sufficient, with minor deficiencies



Manganese (Mn) & Copper (Cu)

Generally sufficient across the district

GIS-based spatial mapping showed clear deficiency zones, highlighting the need for site-specific nutrient management.

Discussion & Significance of the Study

The study confirms that zinc and boron deficiencies are emerging concerns in Pulwama soils. Excessive dependence on chemical fertilizers and limited use of organic inputs have contributed to nutrient imbalance.

Farmer surveys revealed low awareness regarding micronutrients, with most farmers focusing only on NPK fertilizers. Limited soil testing and lack of access to micronutrient fertilizers further aggravate the problem.

The findings emphasize the importance of integrated nutrient management, combining chemical fertilizers with organic amendments, to maintain long-term soil fertility and environmental sustainability.

Conclusion & Recommendations

Conclusion

The study concludes that while Pulwama soils are generally fertile, deficiencies of zinc and boron pose serious risks to sustainable agriculture. Continuous monitoring and balanced nutrient management are essential to prevent further soil degradation and yield decline.

Key Recommendations

01

Regular Soil Testing

Regular soil and leaf nutrient testing

02

Targeted Fertilization

Targeted application of zinc and boron fertilizers

03

Organic Amendments

Increased use of organic manures and compost

04

Integrated Management

Adoption of integrated nutrient management practices

05

Farmer Training

Farmer awareness and training programs

06

Sustainable Practices

Promotion of eco-friendly and sustainable agricultural practices